

A GTMD-First Light-Front Wigner and SCET Framework for Spin-1 Deuteron TMDs

Technical draft of the formalism

Working note

July 8, 2026

Abstract

This note develops a first-principles phenomenological framework for the full leading-power spin-1 partonic structure of the deuteron. The central organizing object is a zero-skewness generalized transverse-momentum-dependent correlator (GTMD), or equivalently its light-front partonic Wigner transform. This GTMD-first viewpoint is useful because the same parent correlator reduces to TMDs in the forward limit, to GPDs after transverse-momentum integration, and to ordinary PDFs and form-factor sum rules after the appropriate further projections. The nuclear part of the model is not supplied by SCET: it is encoded in a light-front spin-density matrix and Wigner/spectral distribution built from realistic deuteron wave functions and supplemented by coherent, exchange-current, shadowing, and non-nucleonic components where needed. Quark, antiquark, and gluon correlators are kept distinct at the operator level, but are assembled into a common partonic vector when constructing measured DIS, SIDIS, Drell-Yan, and gluon-sensitive observables. SCET/TMD factorization supplies Wilson lines, soft subtraction, Collins-Soper evolution, small- b_T matching, and the fixed-order Y term. Inclusive DIS provides the first required normalization check: the tensor-polarized quark distributions must reproduce the spin-1 structure function b_1 . The gluon sector, including double-helicity-flip/gluon-transversity channels, is included as part of the unified spin-1 system, but not used as the primary normalization anchor because present constraints are dominantly inclusive-DIS and SIDIS constraints on quark and antiquark tensor structure.

Contents

1	Scope and guiding principles	3
2	Conventions and kinematics	4
3	Spin-1 polarization and density matrices	5
3.1	Helicity-state tensor difference	6
4	Inclusive DIS and the b_1 anchor	6
5	GTMD-first parent formalism	7
5.1	Why the GTMD layer is the natural starting point	7
5.2	Zero-skewness off-forward kinematics	8

5.3	Quark and antiquark GTMD correlators	8
5.4	Gluon GTMD correlator	8
5.5	Partonic Wigner transform and marginals	9
5.6	Projection from GTMD helicity matrices to the full spin-1 TMD set	9
5.7	GTMD-level nuclear convolution	10
5.8	Constraint map	10
6	Light-front deuteron wave function and Wigner density	11
6.1	Light-front two-nucleon wave function	12
6.2	Momentum density matrix	12
6.3	Transverse Wigner density	13
7	Collinear nuclear convolution and b_1	13
8	Spin-1 TMD operator definitions	14
8.1	Quark TMD correlator	14
8.2	Gluon TMD correlator	15
8.3	SCET representation	15
9	Leading-twist spin-1 TMD inventory	15
10	Tensor-basis construction and projectors	16
11	TMD convolution from the Wigner density	17
11.1	Momentum-space convolution	17
11.2	b_T -space convolution	18
12	Small-b_T OPE and matching	18
13	TMD evolution	19
14	Model ansatz over phase space	20
15	Fourier–Bessel transforms and transverse rank	21
16	SCET factorization for observables	21
16.1	Drell–Yan	21
16.2	SIDIS	21
16.3	Inclusive DIS limit	22
16.4	Gluon-sensitive observables	22
17	Matching low, moderate, and high transverse momentum	22

18 Positivity, sum rules, and consistency tests	22
18.1 Density-matrix positivity	23
18.2 Partonic helicity-density positivity	23
18.3 Collinear and tensor sum rules	23
18.4 Evolution consistency	23
19 Numerical infrastructure	24
20 Comparison with the gluon spectator model	24
21 Development roadmap	25
22 Open derivations and near-term tasks	26
23 Conclusion	27
A Useful helicity combinations	27
B Rank-dependent Bessel kernels	27
C Minimal data and input list	28

1 Scope and guiding principles

The goal is a unified model for spin-1 deuteron TMDs over the practical phase space in

$$x, \quad k_T, \quad b_T, \quad Q, \quad (1)$$

for all partonic species

$$a = q, \bar{q}, g \quad (2)$$

and for all spin-1 target-polarization channels. The intended output is not merely a set of model curves for gluon TMDs, but an infrastructure in which DIS, SIDIS, Drell–Yan, and gluon-sensitive channels can all be evaluated consistently.

The framework has five layers:

$$\begin{array}{lcl}
\boxed{\text{deuteron LF Wigner/spectral density}} & \longrightarrow & \boxed{\text{spin-1 partonic GTMDs/Wigner functions}} \\
& \longrightarrow & \boxed{\text{TMD, GPD, PDF, and form-factor marginals}} \\
& \longrightarrow & \boxed{\text{collinear tensor PDFs and } b_1} \\
& \longrightarrow & \boxed{\text{SCET/TMD-factorized observables and evolution}}.
\end{array} \quad (3)$$

The GTMD layer is introduced to make all reductions and constraints commute: the forward limit gives TMDs, transverse-momentum integration gives GPDs, the forward collinear limit gives PDFs, and the tensor helicity difference gives the inclusive b_1 anchor.

The framework should be able to compare with the spectator-model gluon calculation of Xie, Chen, and Lu [1], but its organizing principle is broader. Their calculation is a useful baseline for leading-twist, time-reversal-even gluon TMDs of a tensor-polarized deuteron. Here the model target is the entire spin-1 TMD system, with the gluon TMDs as one sector.

The main principles are:

- P1. Use b_1 as the inclusive normalization anchor.** The model must reproduce the leading-order parton-model relation between tensor-polarized quark distributions and the spin-1 structure function b_1 [2, 4, 5].
- P2. Separate correlators, unified phenomenology.** Quark and antiquark functions are matrix elements of bilocal quark operators, while gluon functions are matrix elements of bilocal field-strength operators with adjoint Wilson lines. Therefore their microscopic correlators, tensor decompositions, matching coefficients, and hard partonic channels are different. Measured tensor-polarized observables, however, receive weighted sums of quark, antiquark, and gluon channels. The model should therefore store separate operator-level entries but assemble them into a common partonic vector for phenomenology.
- P3. Use SCET as a factorization and evolution layer.** SCET fixes the hard, collinear, and soft separation, Wilson-line structure, soft subtraction, rapidity evolution, and small- b_T matching. It does not determine the nonperturbative deuteron matrix element by itself [12, 13, 14].
- P4. Use light-front nuclear structure for the boundary model.** A deuteron Wigner/spectral density built from LF wave functions supplies Fermi motion, binding, S - D interference, tensor polarization, and spin-density-matrix constraints.
- P5. Separate nucleonic, coherent, exchange, and non-nucleonic mechanisms.** A single impulse-approximation convolution cannot describe the full phase space, especially low x and gluon double-helicity-flip channels.
- P6. Make all conventions explicit.** Spin-1 tensor normalizations vary in the literature. We use convention-invariant helicity differences where possible and provide conversion factors to common S_{LL} conventions.

2 Conventions and kinematics

We use metric $g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ and $\epsilon^{0123} = +1$. Light-front components are

$$v^\pm = \frac{v^0 \pm v^3}{\sqrt{2}}, \quad v \cdot w = v^+ w^- + v^- w^+ - \mathbf{v}_T \cdot \mathbf{w}_T. \quad (4)$$

The lightlike basis vectors satisfy

$$n^2 = \bar{n}^2 = 0, \quad n \cdot \bar{n} = 1, \quad v^+ = n \cdot v, \quad v^- = \bar{n} \cdot v, \quad (5)$$

and

$$v^\mu = v^+ \bar{n}^\mu + v^- n^\mu + v_T^\mu. \quad (6)$$

The transverse metric and antisymmetric tensor are

$$g_T^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n^\nu - n^\mu \bar{n}^\nu, \quad \epsilon_T^{\mu\nu} = \epsilon^{\mu\nu\alpha\beta} \bar{n}_\alpha n_\beta. \quad (7)$$

In two-dimensional Euclidean transverse notation we use δ_T^{ij} and ϵ_T^{ij} with $\epsilon_T^{12} = +1$.

The deuteron momentum is chosen mostly plus-collinear,

$$P_D^\mu = P_D^+ \bar{n}^\mu + \frac{M_D^2}{2P_D^+} n^\mu, \quad P_{DT} = 0. \quad (8)$$

Bjorken x is $x_B = Q^2/(2P_D \cdot q)$ in inclusive DIS, while the partonic light-front fraction is $x = k^+/P_D^+$ for deuteron TMDs. For a nucleon inside the deuteron we use

$$y = \frac{p_N^+}{P_D^+}, \quad 0 < y < 1, \quad (9)$$

or, equivalently, the nuclear convention $\alpha = 2y$. We use y throughout the formal derivation and note conversion to α where useful.

For TMD observables with a hard scale Q and measured transverse momentum $q_T \ll Q$, define

$$\lambda \sim \frac{q_T}{Q} \ll 1. \quad (10)$$

The leading-power SCET_{II} mode scalings for a two-direction process are

$$p_n^\mu \sim Q(1, \lambda^2, \lambda), \quad p_{\bar{n}}^\mu \sim Q(\lambda^2, 1, \lambda), \quad p_s^\mu \sim Q(\lambda, \lambda, \lambda), \quad (11)$$

where entries denote $(+, -, T)$. Collinear and soft modes have the same virtuality but different rapidity. This is why TMD factorization requires a rapidity regulator or an equivalent Collins–Soper scale [15, 16, 17, 18].

3 Spin-1 polarization and density matrices

A spin-1 target has a vector polarization and an independent rank-2 tensor polarization. Covariantly,

$$S \cdot P_D = 0, \quad T^{\mu\nu} = T^{\nu\mu}, \quad T^\mu_\mu = 0, \quad P_{D\mu} T^{\mu\nu} = 0. \quad (12)$$

The spin-density matrix for a massive spin-1 state can be represented by attaching polarization vectors to the 3×3 helicity density matrix,

$$\rho^{\alpha\beta}(P_D) = \sum_{\Lambda, \Lambda'=-1}^{+1} \rho_{\Lambda\Lambda'} \epsilon^\alpha(P_D, \Lambda) \epsilon^{*\beta}(P_D, \Lambda'). \quad (13)$$

A common covariant decomposition is

$$\rho^{\alpha\beta} = \frac{1}{3} \left(-g^{\alpha\beta} + \frac{P_D^\alpha P_D^\beta}{M_D^2} \right) + \frac{i}{2M_D} \epsilon^{\alpha\beta\gamma\delta} P_{D\gamma} S_\delta - T^{\alpha\beta}, \quad (14)$$

up to sign conventions for $T^{\alpha\beta}$. Equation (14) is a convention statement; all later formulae are expressed in terms of helicity combinations or explicitly stated S_{LL}, S_{LT}, S_{TT} variables.

In the deuteron rest frame, the tensor polarization can be constructed from spin-1 matrices J^i :

$$S^i = \text{Tr}(\rho J^i), \quad T^{ij} = \frac{1}{2} \text{Tr}[\rho \{J^i, J^j\}] - \frac{2}{3} \delta^{ij} \text{Tr} \rho. \quad (15)$$

The five independent tensor components are conventionally organized as

$$S_{LL}, \quad S_{LT}^i, \quad S_{TT}^{ij} = S_{TT}^{ji}, \quad S_{TT}^{ii} = 0, \quad (16)$$

where LL is a scalar under transverse rotations, LT is a transverse vector, and TT is a symmetric traceless transverse tensor.

3.1 Helicity-state tensor difference

For inclusive DIS and for model normalization it is safest to define the tensor-polarized difference directly in terms of deuteron helicity states:

$$\delta_T F \equiv F^{\Lambda=0} - \frac{1}{2} \left(F^{\Lambda=+1} + F^{\Lambda=-1} \right). \quad (17)$$

This definition is independent of whether a paper normalizes S_{LL} as $S_{LL}^0 = 1$ or $S_{LL}^0 = -1$. If one writes

$$F^\Lambda = F_U + S_{LL}^\Lambda F_{LL} \quad (18)$$

with the common choice

$$S_{LL}^{\Lambda=0} = -1, \quad S_{LL}^{\Lambda=\pm 1} = \frac{1}{2}, \quad (19)$$

then

$$\delta_T F = -\frac{3}{2} F_{LL}, \quad F_{LL} = -\frac{2}{3} \delta_T F. \quad (20)$$

With the opposite convention $S_{LL}^0 = +1$, $S_{LL}^\pm = -1/2$, the sign in Eq. (20) flips. In this draft, all b_1 constraints are imposed on $\delta_T q$ and $\delta_T \bar{q}$; conversion to f_{1LL} is done only after a tensor-polarization convention is chosen.

4 Inclusive DIS and the b_1 anchor

A spin-1 target has additional inclusive DIS structure functions not present for a spin-1/2 target. At leading twist, the tensor structure functions are b_1 and b_2 , with the Callan–Gross-like relation

$$b_2(x, Q^2) = 2x b_1(x, Q^2) \quad (21)$$

in the Bjorken scaling limit [2, 5]. Higher-twist tensor functions b_3 and b_4 can enter at finite Q^2 but are not part of the leading-power TMD boundary model.

For a fixed deuteron helicity Λ , the leading-order parton-model relation for the spin-averaged DIS structure function is

$$F_1^\Lambda(x, Q^2) = \frac{1}{2} \sum_q e_q^2 \left[q_D^\Lambda(x, Q^2) + \bar{q}_D^\Lambda(x, Q^2) \right]. \quad (22)$$

Taking the tensor combination (17), one obtains

$$\begin{aligned} b_1^D(x, Q^2) &= F_1^{\Lambda=0}(x, Q^2) - \frac{1}{2} \left[F_1^{\Lambda=+1}(x, Q^2) + F_1^{\Lambda=-1}(x, Q^2) \right] \\ &= \frac{1}{2} \sum_q e_q^2 \left[\delta_T q_D(x, Q^2) + \delta_T \bar{q}_D(x, Q^2) \right] + \mathcal{O}(\alpha_s, 1/Q^2). \end{aligned} \quad (23)$$

Equation (23) is the first hard constraint on the model. In the TMD model it becomes

$$\delta_T q_D(x, Q^2) = \int d^2 \mathbf{k}_T \delta_T f_{q/D}(x, \mathbf{k}_T; Q^2) \quad (24)$$

with the appropriate soft-subtracted TMD definition. In a convention using $f_{1LL}^{q/D}$ with $S_{LL}^0 = -1$, Eq. (20) gives

$$\int d^2 \mathbf{k}_T f_{1LL}^{q/D}(x, \mathbf{k}_T; Q^2) = -\frac{2}{3} \delta_T q_D(x, Q^2). \quad (25)$$

This factor is frequently a source of sign and normalization mistakes.

The HERMES measurement established that b_1^D is nonzero in the measured kinematic range [4]. Standard deuteron convolution calculations have shown tension with those data, motivating mechanisms beyond a minimal impulse approximation [5]. Our framework therefore treats b_1 not merely as a check but as a diagnostic that determines how much non-impulse physics must be added.

5 GTMD-first parent formalism

5.1 Why the GTMD layer is the natural starting point

The most economical object for a full spin-1 model is not a TMD by itself but a generalized transverse-momentum-dependent correlator (GTMD), the off-forward parent correlator underlying partonic Wigner distributions, TMDs, and GPDs [22, 23, 24]. At zero skewness, a GTMD depends on

$$x, \quad \mathbf{k}_T, \quad \Delta_T, \quad (26)$$

where Δ_T is the transverse momentum transfer to the deuteron. Fourier transformation in Δ_T gives a partonic light-front Wigner distribution in transverse phase space. The reason to introduce this object at the beginning is that it contains, as limiting cases, the distributions against which the model can be constrained:

$$\begin{aligned} \mathbb{W}^{a/D}(x, \mathbf{k}_T, \Delta_T) &\xrightarrow{\Delta_T \rightarrow 0} \Phi_{\text{TMD}}^{a/D}(x, \mathbf{k}_T) \xrightarrow{\int d^2 \mathbf{k}_T} f^{a/D}(x), \\ &\downarrow \int d^2 \mathbf{k}_T \\ H^{a/D}(x, 0, -\Delta_T^2) &\xrightarrow{\int dx} \text{deuteron form factors.} \end{aligned} \quad (27)$$

Here $a = q, \bar{q}, g$, and the spin, Dirac/Lorentz, and gauge-link labels on $\mathbb{W}$ are suppressed. Equation (27) is the basic consistency diagram for the model. It says that a TMD model should not be built in isolation if one wants to use all available nuclear and partonic constraints. The same parent object should reduce to the tensor PDFs entering b_1 , to the TMDs entering SIDIS and Drell–Yan, and to zero-skewness GPDs whose moments reproduce charge, magnetic, and quadrupole form-factor constraints.

This note therefore uses the following hierarchy:

$$\begin{aligned} &\boxed{\text{operator-level GTMD}} \implies \boxed{\text{partonic Wigner distribution}} \\ &\implies \boxed{\text{TMD/GPD/PDF marginals}} \implies \boxed{\text{SCET-factorized observables}}. \end{aligned} \quad (28)$$

The phrase “GTMD from the Wigner function” should be interpreted carefully. In QCD, the GTMD and the partonic light-front Wigner distribution are Fourier conjugates in Δ_T and transverse position. In the nuclear model, the deuteron light-front Wigner/spectral density supplies the nonperturbative convolution kernel for the deuteron’s partonic GTMD. These are related ideas but not the same mathematical object: one is a partonic QCD correlator; the other is a nuclear phase-space density for nucleonic degrees of freedom. This distinction mirrors the lesson from nuclear Wigner studies: the Wigner function is a quasi-density that links coordinate and momentum information but should not be interpreted as a positive probability density point by point [25].

5.2 Zero-skewness off-forward kinematics

Let

$$p = P - \frac{\Delta}{2}, \quad p' = P + \frac{\Delta}{2}, \quad (29)$$

where p and p' are the incoming and outgoing deuteron momenta. The skewness is

$$\xi = -\frac{\Delta^+}{2P^+}. \quad (30)$$

The initial implementation should use

$$\xi = 0, \quad \Delta^+ = 0, \quad \Delta^\mu = (0, 0, \mathbf{\Delta}_T), \quad t = \Delta^2 = -\mathbf{\Delta}_T^2. \quad (31)$$

This is sufficient for the TMD limit, the transverse Wigner transform, and zero-skewness GPD/form-factor constraints. A future extension to $\xi \neq 0$ is required for full GPD polynomiality tests, but it is not necessary for the first TMD phenomenology layer.

The parton average transverse momentum is $\mathbf{k}_T$. The Fourier-conjugate transverse position is denoted $\mathbf{b}_\Delta$ when it is conjugate to $\mathbf{\Delta}_T$. This should be distinguished from the TMD impact parameter $\mathbf{b}_T$ conjugate to measured transverse momentum. In many formulas both are denoted by b_T in the literature. The implementation should store them separately, for example as `bDelta` and `bTMD`, to avoid mixing the transverse imaging and TMD-evolution Fourier transforms.

5.3 Quark and antiquark GTMD correlators

For a quark flavor q , define the unsubtracted zero-skewness quark GTMD correlator

$$\begin{aligned} \mathbb{W}_{\Lambda'\Lambda}^{q[\Gamma],\eta}(x, \mathbf{k}_T, \mathbf{\Delta}_T) &= \frac{1}{2} \int \frac{dz^- d^2 \mathbf{z}_T}{(2\pi)^3} \exp\left(ixP^+ z^- - i\mathbf{k}_T \cdot \mathbf{z}_T\right) \\ &\times \langle p', \Lambda' | \bar{\psi}_q\left(-\frac{z}{2}\right) \mathcal{W}_\eta\left[-\frac{z}{2}, \frac{z}{2}\right] \Gamma \psi_q\left(\frac{z}{2}\right) | p, \Lambda \rangle \Big|_{z^+=0}. \end{aligned} \quad (32)$$

Here

$$\Gamma \in \{\gamma^+, \gamma^+ \gamma_5, i\sigma^{i+} \gamma_5\} \quad (33)$$

at leading twist. The staple direction and color representation of the Wilson line are encoded in η and in any additional process labels. Antiquark GTMDs may be represented either by the negative- x support of the quark correlator or by independent positive- x distributions. For phenomenology we store q and $\bar{q}$ separately, while remembering that they come from the same quark operator definition.

The correlator in Eq. (32) is a 3×3 matrix in deuteron helicity space. After projecting onto quark helicity or quark transverse polarization, it can also be viewed as a joint parton-target helicity-density matrix. This is the object from which all quark spin-1 TMDs are projected.

5.4 Gluon GTMD correlator

For gluons, the corresponding unsubtracted zero-skewness correlator is

$$\mathbb{W}_{\Lambda'\Lambda}^{g,ij,\eta}(x, \mathbf{k}_T, \mathbf{\Delta}_T) = \frac{1}{xP^+} \int \frac{dz^- d^2 \mathbf{z}_T}{(2\pi)^3} \exp\left(ixP^+ z^- - i\mathbf{k}_T \cdot \mathbf{z}_T\right)$$

$$\times \langle p', \Lambda' | F_a^{+i} \left(-\frac{z}{2} \right) \mathcal{W}_\eta^{ab} \left[-\frac{z}{2}, \frac{z}{2} \right] F_b^{+j} \left(\frac{z}{2} \right) | p, \Lambda \rangle \Big|_{z^+=0}. \quad (34)$$

The indices i, j are transverse. The trace, antisymmetric, and symmetric-traceless parts of $\mathbb{W}^{g,ij}$ correspond, respectively, to unpolarized gluons, circularly polarized gluons, and linearly polarized/double-helicity-flip gluon structures. The target spin-1 density matrix then supplies the U, L, T, LL, LT, TT target-polarization projections. Gluon GTMDs must retain a gauge-link/color-flow label because different processes can probe different Wilson-line structures.

Equations (32) and (34) are distinct operator definitions. They should not be collapsed into a single microscopic correlator. The unification occurs only after each sector has been projected onto a common phenomenological basis of functions and inserted into observable factorization formulae.

5.5 Partonic Wigner transform and marginals

The transverse partonic Wigner distribution is the Fourier transform

$$\mathcal{R}_{\Lambda'\Lambda}^{a[J],\eta}(x, \mathbf{k}_T, \mathbf{b}_\Delta) = \int \frac{d^2 \Delta_T}{(2\pi)^2} e^{-i\Delta_T \cdot \mathbf{b}_\Delta} \mathbb{W}_{\Lambda'\Lambda}^{a[J],\eta}(x, \mathbf{k}_T, \Delta_T), \quad (35)$$

where J denotes the quark Dirac or gluon Lorentz projection. This object is a light-front phase-space quasi-distribution for partons in a spin-1 target. Its basic reductions are

$$\Phi_{\Lambda'\Lambda}^{a[J],\eta}(x, \mathbf{k}_T) = \mathbb{W}_{\Lambda'\Lambda}^{a[J],\eta}(x, \mathbf{k}_T, \Delta_T = 0), \quad (36)$$

$$H_{\Lambda'\Lambda}^{a[J]}(x, 0, -\Delta_T^2) = \int d^2 \mathbf{k}_T \mathbb{W}_{\Lambda'\Lambda}^{a[J]}(x, \mathbf{k}_T, \Delta_T), \quad (37)$$

$$f_{\Lambda\Lambda}^{a[J]}(x) = \int d^2 \mathbf{k}_T \mathbb{W}_{\Lambda\Lambda}^{a[J]}(x, \mathbf{k}_T, 0). \quad (38)$$

Equation (36) is the TMD limit. Equation (37) is the zero-skewness GPD limit. Equation (38) is the collinear PDF limit. For tensor polarization,

$$\delta_T f_D^{a[J]}(x) = \int d^2 \mathbf{k}_T \left[\mathbb{W}_{00}^{a[J]}(x, \mathbf{k}_T, 0) - \frac{1}{2} \left(\mathbb{W}_{++}^{a[J]}(x, \mathbf{k}_T, 0) + \mathbb{W}_{--}^{a[J]}(x, \mathbf{k}_T, 0) \right) \right]. \quad (39)$$

For $J = \gamma^+$ and $a = q, \bar{q}$, Eq. (39) is the input entering the b_1 relation in Eq. (23).

Local-current and energy-momentum-tensor constraints appear as x moments of the GPD limit. Schematically,

$$\int dx H_{\Lambda'\Lambda}^{q[V]}(x, 0, t) \longrightarrow \langle p', \Lambda' | J_q^+(0) | p, \Lambda \rangle, \quad (40)$$

$$\sum_{a=q,\bar{q},g} \int dx x H_{\Lambda'\Lambda}^{a[V]}(x, 0, t) \longrightarrow \langle p', \Lambda' | T_a^{++}(0) | p, \Lambda \rangle. \quad (41)$$

For a spin-1 target these moments constrain charge, magnetic, quadrupole, and gravitational form-factor combinations. These constraints are not replacements for b_1 ; they are independent validation tests of the off-forward nuclear kernel.

5.6 Projection from GTMD helicity matrices to the full spin-1 TMD set

For each species a define a parton-target helicity matrix

$$\mathcal{M}_{\lambda'_a \Lambda'; \lambda_a \Lambda}^a(x, \mathbf{k}_T, \Delta_T; \eta), \quad (42)$$

where λ_a labels quark helicity for quarks and gluon helicity for gluons. The leading-twist TMDs are obtained by setting $\Delta_T = 0$ and tracing this matrix with parton-polarization projectors and spin-1 target projectors:

$$F_A^{a/D}(x, \mathbf{k}_T; \eta) = \text{Tr}_{\lambda_a, \Lambda} \left[P_A^{(a)}(\hat{k}_T, S, T) \mathcal{M}^a(x, \mathbf{k}_T, 0; \eta) \right]. \quad (43)$$

Here A labels the named TMD, for example f_1^q , g_{1L}^q , h_1^q , f_{1LL}^q , $h_{1LL}^{\perp q}$, f_{1LT}^g , or h_{1TT}^g . Equation (43) is the formal basis for a machine-readable TMD registry. The registry should store

(a , J , η , target channel, parton polarization, transverse rank, collinear limit, known matching status). (44)

This ensures that quark, antiquark, and gluon functions remain separate as correlators but are available together for measured phenomenology.

5.7 GTMD-level nuclear convolution

The impulse-approximation model should be formulated at the GTMD level and only then reduced to TMDs, GPDs, or PDFs. At zero skewness, the most general one-body nucleonic contribution has the schematic form

$$\begin{aligned} \mathbb{W}_{\Lambda'\Lambda}^{a/D, \text{IA}}(x, \mathbf{k}_T, \Delta_T; \eta) = & \sum_{N=p,n} \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} \sum_{\lambda_N, \lambda'_N} \mathcal{S}_{\Lambda'\Lambda; \lambda'_N \lambda_N}^{N/D}(y, \mathbf{p}_T, \Delta_T) \\ & \times \mathbb{W}_{\lambda'_N \lambda_N}^{a/N, \eta} \left(\frac{x}{y}, \mathbf{k}_T - \frac{x}{y} \mathbf{p}_T, \Delta_{T,N} \right) \mathcal{J}_G(y, \mathbf{p}_T, \Delta_T) + \mathbb{W}_{\Lambda'\Lambda}^{a/D, \text{coh}} + \mathbb{W}_{\Lambda'\Lambda}^{a/D, \text{exch}} + \mathbb{W}_{\Lambda'\Lambda}^{a/D, \text{nonN}}. \end{aligned} \quad (45)$$

Here $\mathcal{S}^{N/D}$ is the off-forward light-front deuteron spectral density, including nucleon spin indices. In a symmetric transverse frame one convenient overlap definition is

$$\mathcal{S}_{\Lambda'\Lambda; \lambda'_N \lambda_N}^{N/D}(y, \mathbf{p}_T, \Delta_T) = \sum_{\lambda_s} \Psi_{\Lambda'}^{* \lambda'_N \lambda_s} \left(y, \mathbf{p}_T + (1-y) \frac{\Delta_T}{2} \right) \Psi_{\Lambda}^{\lambda_N \lambda_s} \left(y, \mathbf{p}_T - (1-y) \frac{\Delta_T}{2} \right), \quad (46)$$

up to the Jacobian and spin-rotation conventions chosen for the LF wave function. The variable $\Delta_{T,N}$ is the transverse momentum transfer seen by the active nucleon. In the simplest one-body convention $\Delta_{T,N} = \Delta_T$; in a fully consistent constituent overlap it may carry y -dependent recoil factors. This is not a harmless detail: the chosen mapping must be validated by the TMD limit, the GPD/form-factor limit, and baryon-number normalization.

The forward limit of Eq. (45) gives the TMD convolution in Sec. 11. The $\mathbf{k}_T$ -integrated limit gives the usual light-front convolution for deuteron GPDs at zero skewness. The double integral over x and $\mathbf{k}_T$ gives deuteron form-factor constraints. These reductions should be used as unit tests of the formalism.

5.8 Constraint map

The GTMD-first model should use every available constraint, but each constraint acts on a specific marginal or moment. The following table summarizes the constraint logic.

Table 1: Constraint map for the GTMD-first spin-1 deuteron model.

Constraint	GTMD reduction	Model role
Inclusive F_1, F_2	$\Delta_T = 0$, integrate $\mathbf{k}_T$	Normalizes unpolarized quark and antiquark PDFs.
Tensor structure function b_1	Helicity difference of γ^+ quark PDF limit	Primary tensor-polarized quark/antiquark normalization anchor.
Leading-twist $b_2 = 2xb_1$	Same collinear tensor PDF limit	Checks leading-twist normalization and target-mass/higher-twist separation.
Deuteron electromagnetic form factors	Integrate x and $\mathbf{k}_T$ of vector GPD limit	Constrains off-forward spin-1 density, especially quadrupole structure.
Momentum sum rule	x moment of quark+gluon forward limit	Couples quark, antiquark, and gluon sectors.
Nucleon PDFs and TMDs	Inputs to $\mathbb{W}^{a/N}$ and its TMD/PDF limits	Supplies proton/neutron partonic baseline in the impulse term.
SIDIS tensor asymmetries	$\Delta_T = 0$ TMD limit with fragmentation	First practical constraints on f_{1LL} and related tensor TMDs.
Drell–Yan tensor asymmetries	$\Delta_T = 0$ TMD limit in quark-antiquark channels	Constrains tensor-polarized antiquarks and sign-changing gauge links.
Gluon-sensitive final states	Gluon TMD limit of $\mathbb{W}^{g,ij}$	Constrains gluon tensor TMDs where factorization is established.
Small- b_T OPE	Perturbative TMD limit at small TMD impact parameter	Enforces correct perturbative tail and quark-gluon matching.
Hermiticity, parity, time reversal	Operator-level GTMD relations	Reduces independent functions and checks signs/conventions.
Positivity	Forward TMD helicity matrix	Semidefinite bounds on allowed fitted functions.

This table is also a development roadmap: first make the GTMD reductions reproduce unpolarized PDFs, form factors, and b_1 ; then turn on the TMD-specific transverse-momentum dependence and SCET evolution; then add the less constrained gluon and off-diagonal tensor sectors.

6 Light-front deuteron wave function and Wigner density

The GTMD definitions in Sec. 5 are QCD operator definitions. The present section supplies the nuclear part of the boundary model: a light-front two-nucleon density matrix and its Wigner/spectral representation. This nuclear Wigner density is the relativistic analogue of the phase-space Wigner function used in deuteron short-range-correlation studies. Neff and Feldmeier, for example, define a spin-space Wigner matrix for the deuteron whose coordinate and momentum marginals reproduce the corresponding densities, and emphasize that its negative regions are quantum interference effects rather than probabilities [25]. We use the same phase-space logic, but translated to light-front

variables appropriate for DIS and TMD factorization.

There are therefore two Wigner-like objects in the formalism:

$$W_{\Lambda'\Lambda}^{N/D}(y, \mathbf{p}_T, \mathbf{R}_T) \quad \text{nuclear LF Wigner/spectral density for a nucleon in the deuteron,} \quad (47)$$

$$\mathcal{R}_{\Lambda'\Lambda}^{a[J]/D}(x, \mathbf{k}_T, \mathbf{b}_\Delta) \quad \text{partonic LF Wigner distribution obtained from a GTMD.} \quad (48)$$

The first is a model input built from deuteron wave functions; the second is a QCD parton correlator. The impulse approximation convolves the two levels through Eq. (45).

6.1 Light-front two-nucleon wave function

The leading nucleonic component of the deuteron is represented by a light-front wave function

$$\Psi_{\Lambda}^{\lambda_p \lambda_n}(y, \mathbf{p}_T), \quad (49)$$

where Λ is the deuteron helicity, λ_p, λ_n are proton and neutron light-front helicities, $y = p_p^+/P_D^+$ for the proton, and the neutron has fraction $1 - y$ and transverse momentum $-\mathbf{p}_T$. A convenient normalization is

$$\sum_{\lambda_p, \lambda_n} \int_0^1 \frac{dy}{2y(1-y)} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^3} \left| \Psi_{\Lambda}^{\lambda_p \lambda_n}(y, \mathbf{p}_T) \right|^2 = 1, \quad (50)$$

though other normalizations are possible. The normalization must be fixed once and propagated through all convolution formulae.

To connect with standard deuteron wave-function theory, introduce the internal three-momentum $\mathbf{k} = (\mathbf{p}_T, k_z)$ by

$$y = \frac{E_k + k_z}{2E_k}, \quad E_k = \sqrt{m_N^2 + \mathbf{k}^2}, \quad k_z = \left(y - \frac{1}{2}\right) 2E_k. \quad (51)$$

The instant-form rest-frame wave function is

$$\Phi_{\Lambda}^{\sigma_p \sigma_n}(\mathbf{k}) = \sum_{L=0,2} \sum_{m_L, m_S} \langle L m_L, 1 m_S | 1 \Lambda \rangle \langle \frac{1}{2} \sigma_p, \frac{1}{2} \sigma_n | 1 m_S \rangle Y_{L m_L}(\hat{\mathbf{k}}) u_L(k), \quad (52)$$

where $u_0(k)$ and $u_2(k)$ are the S - and D -wave radial components. The light-front wave function is obtained by applying the appropriate canonical-to-light-front spin rotations and a Jacobian factor:

$$\Psi_{\Lambda}^{\lambda_p \lambda_n}(y, \mathbf{p}_T) = \sqrt{\frac{\partial k_z}{\partial y}} \sum_{\sigma_p, \sigma_n} D_{\lambda_p \sigma_p}^{1/2}[R_M(p_p)] D_{\lambda_n \sigma_n}^{1/2}[R_M(p_n)] \Phi_{\Lambda}^{\sigma_p \sigma_n}(\mathbf{k}). \quad (53)$$

Equation (53) is schematic; the exact spin rotation depends on the chosen light-front quantum mechanics convention. The model code should support several deuteron wave-function inputs, for example AV18 [20], CD-Bonn [21], or chiral EFT wave functions. Differences among these inputs become part of the nuclear-structure uncertainty.

6.2 Momentum density matrix

The one-body LF momentum density matrix for finding a nucleon $N = p, n$ in a deuteron with helicity transition $\Lambda \rightarrow \Lambda'$ is

$$\rho_{\Lambda'\Lambda}^N(y, \mathbf{p}_T) = \sum_{\lambda_N, \lambda_s} \Psi_{\Lambda'}^{*\lambda_N \lambda_s}(y, \mathbf{p}_T) \Psi_{\Lambda}^{\lambda_N \lambda_s}(y, \mathbf{p}_T), \quad (54)$$

where λ_s denotes the spectator nucleon's helicity. For the neutron distribution, y and $\mathbf{p}_T$ refer to the neutron and the proton is the spectator. Equation (54) is positive semidefinite in deuteron helicity space at each (y, p_T) when the full helicity information is retained.

The unpolarized and longitudinal tensor combinations are

$$\rho_U^N(y, \mathbf{p}_T) = \frac{1}{3} \left[\rho_{++}^N + \rho_{00}^N + \rho_{--}^N \right], \quad (55)$$

$$\delta_T \rho^N(y, \mathbf{p}_T) = \rho_{00}^N - \frac{1}{2} \left(\rho_{++}^N + \rho_{--}^N \right). \quad (56)$$

The off-diagonal components generate LT and TT tensor polarizations. A convenient spherical-tensor decomposition is

$$\rho_{\Lambda'\Lambda}^N = \sum_{K=0}^2 \sum_{M=-K}^K \rho_{KM}^N \langle 1\Lambda, KM | 1\Lambda' \rangle, \quad (57)$$

where $K = 0$ is unpolarized, $K = 1$ is vector polarization, and $K = 2$ is tensor polarization. The $M = 0$ member of $K = 2$ is the LL channel, $M = \pm 1$ maps to LT , and $M = \pm 2$ maps to TT .

6.3 Transverse Wigner density

For TMDs and spatial imaging, we need a transverse phase-space density. Define a LF Wigner distribution by Fourier transforming the off-forward density in transverse momentum transfer $\mathbf{\Delta}_T$:

$$W_{\Lambda'\Lambda}^N(y, \mathbf{p}_T, \mathbf{R}_T) = \int \frac{d^2 \mathbf{\Delta}_T}{(2\pi)^2} e^{-i \mathbf{\Delta}_T \cdot \mathbf{R}_T} \rho_{\Lambda'\Lambda}^N(y, \mathbf{p}_T, \mathbf{\Delta}_T), \quad (58)$$

$$\rho_{\Lambda'\Lambda}^N(y, \mathbf{p}_T, \mathbf{\Delta}_T) = \sum_{\lambda_N, \lambda_s} \Psi_{\Lambda'}^{*\lambda_N \lambda_s} \left(y, \mathbf{p}_T + (1-y) \frac{\mathbf{\Delta}_T}{2} \right) \Psi_{\Lambda}^{\lambda_N \lambda_s} \left(y, \mathbf{p}_T - (1-y) \frac{\mathbf{\Delta}_T}{2} \right). \quad (59)$$

The factor $(1-y)$ in Eq. (59) is the symmetric-frame overlap factor for the active constituent. If the active constituent is the other nucleon or if a different frame convention is used, the shift factors must be changed consistently. The integrated Wigner function returns the momentum density,

$$\int d^2 \mathbf{R}_T W_{\Lambda'\Lambda}^N(y, \mathbf{p}_T, \mathbf{R}_T) = \rho_{\Lambda'\Lambda}^N(y, \mathbf{p}_T). \quad (60)$$

The Wigner function is a quasi-density, not a pointwise probability distribution. Positivity should therefore be imposed at the density-matrix level before Wigner transformation and at the final TMD helicity-density matrix level after convolution.

7 Collinear nuclear convolution and b_1

In impulse approximation, the deuteron matrix element of a one-body partonic operator is approximated by a convolution of the LF nucleon density and the corresponding proton or neutron parton distribution. For a partonic species $a = q, \bar{q}, g$ and deuteron helicity Λ ,

$$f_{a/D}^{\Lambda, \text{IA}}(x; \mu) = \sum_{N=p,n} \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} \rho_{\Lambda\Lambda}^N(y, \mathbf{p}_T) f_{a/N} \left(\frac{x}{y}; \mu \right) \mathcal{J}_f(y, \mathbf{p}_T) \quad (61)$$

where $\mathcal{J}_f$ is a normalization/flux factor fixed by the wave-function convention. In the simplest normalized model $\mathcal{J}_f = 1$; in a fully relativistic implementation it may include factors of y , $1 - y$, and the internal mass M_0 . The same factor must be used in unpolarized and tensor-polarized channels so that baryon number and momentum sum rules can be checked.

The tensor-polarized collinear distribution is

$$\begin{aligned}\delta_T f_{a/D}^{\text{IA}}(x; \mu) &= f_{a/D}^{\Lambda=0, \text{IA}}(x; \mu) - \frac{1}{2} \left[f_{a/D}^{\Lambda=+1, \text{IA}}(x; \mu) + f_{a/D}^{\Lambda=-1, \text{IA}}(x; \mu) \right] \\ &= \sum_{N=p,n} \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} \delta_T \rho^N(y, \mathbf{p}_T) f_{a/N}\left(\frac{x}{y}; \mu\right) \mathcal{J}_f(y, \mathbf{p}_T).\end{aligned}\quad (62)$$

This expression produces a nonzero b_1 when the deuteron tensor density $\delta_T \rho^N$ is nonzero, primarily through S - D interference and D -wave effects in the nucleonic component. The comparison with HERMES and future Jefferson Lab data determines whether this impulse-approximation contribution is sufficient.

We therefore write the full collinear tensor PDF as

$$\delta_T f_{a/D} = \delta_T f_{a/D}^{\text{IA}} + \delta_T f_{a/D}^{\text{coh}} + \delta_T f_{a/D}^{\text{exch}} + \delta_T f_{a/D}^{\text{nonN}}. \quad (63)$$

The terms have the following meanings:

- IA: one-body nucleonic impulse approximation.
- coh: coherent multi-nucleon scattering and shadowing, expected to become important at small x .
- exch: meson-exchange currents and nuclear binding mechanisms not captured by a one-body nucleon PDF.
- nonN: hidden-color, six-quark, or other non-nucleonic components.

The decomposition is additive at the level of matrix elements. It should not be interpreted as a probability decomposition unless the components are represented in an orthogonal Fock-space basis.

8 Spin-1 TMD operator definitions

8.1 Quark TMD correlator

The unsubtracted quark TMD correlator is

$$\begin{aligned}\Phi_{q/D}^{[\Gamma], \text{unsub}}(x, \mathbf{k}_T; S, T) &= \int \frac{d\xi^- d^2 \xi_T}{(2\pi)^3} e^{ixP_D^+ \xi^- - i\mathbf{k}_T \cdot \xi_T} \\ &\times \langle D(P_D, S, T) | \bar{\psi}_q(0) \mathcal{W}_{\text{proc}}[0, \xi] \Gamma \psi_q(\xi) | D(P_D, S, T) \rangle \Big|_{\xi^+ = 0}.\end{aligned}\quad (64)$$

Here $\Gamma \in \{\gamma^+, \gamma^+ \gamma_5, i\sigma^{i+} \gamma_5\}$ at leading twist, and $\mathcal{W}_{\text{proc}}$ is the process-dependent Wilson-line staple. The properly defined TMD is UV-renormalized and soft-subtracted. In impact-parameter space we write schematically

$$\widetilde{\Phi}_{q/D}^{[\Gamma]}(x, \mathbf{b}_T; \mu, \zeta) = Z_q(\mu, \zeta) \Phi_{q/D}^{[\Gamma], \text{unsub}}(x, \mathbf{b}_T; \text{reg}) [S_q(\mathbf{b}_T; \text{reg})]^{-1/2}, \quad (65)$$

where the placement of the square-root soft factor is scheme-dependent. Equation (65) is meant as a scheme declaration template; an implementation must choose a Collins, EIS, RRG, or equivalent scheme and remain consistent.

8.2 Gluon TMD correlator

The full-QCD gluon TMD correlator is

$$\Gamma_{g/D}^{ij,\text{unsub}}(x, \mathbf{k}_T; S, T) = \frac{1}{xP_D^+} \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{ixP_D^+\xi^- - i\mathbf{k}_T \cdot \xi_T} \times \langle D(P_D, S, T) | F_a^{+i}(0) \mathcal{W}_{ab}^{\text{proc}}[0, \xi] F_b^{+j}(\xi) | D(P_D, S, T) \rangle \Big|_{\xi^+=0}. \quad (66)$$

The subtracted object is

$$\tilde{\Gamma}_{g/D}^{ij}(x, \mathbf{b}_T; \mu, \zeta) = Z_g(\mu, \zeta) \Gamma_{g/D}^{ij,\text{unsub}}(x, \mathbf{b}_T; \text{reg}) [S_g(\mathbf{b}_T; \text{reg})]^{-1/2}. \quad (67)$$

The soft function is in the adjoint representation for gluon TMDs. Different gluon gauge-link structures, commonly called *f*-type/Weizsaecker–Williams and *d*-type/dipole structures, correspond to different color flow and process dependence [9]. Therefore every gluon TMD in the model must carry a gauge-link label:

$$\tilde{\Gamma}_{g/D}^{ij} \rightarrow \tilde{\Gamma}_{g/D}^{ij[U,U']}, \quad F_{g/D} \rightarrow F_{g/D}^{[U,U']}. \quad (68)$$

For T-even exploratory work one may start with a fixed $[+, +]$ or $[-, -]$ convention, but phenomenology must match the Wilson-line structure of the chosen process.

8.3 SCET representation

At leading power, the gauge-invariant SCET collinear quark and gluon building blocks are

$$\chi_n(x) = W_n^\dagger(x) \xi_n(x), \quad \mathcal{B}_{n\perp}^{\mu a}(x) = \frac{1}{g} \left[W_n^\dagger i D_{n\perp}^\mu W_n \right]^a(x). \quad (69)$$

The leading gluon TMD operator may be written schematically as

$$\mathcal{O}_g^{ij}(b) = \mathcal{B}_{n\perp}^{ia}(b) \mathcal{S}_{ab}^{\text{proc}}(b, 0) \mathcal{B}_{n\perp}^{jb}(0), \quad (70)$$

where $\mathcal{S}_{ab}^{\text{proc}}$ denotes the soft Wilson-line structure that remains after the BPS field redefinition. The deuteron TMD is the deuteron matrix element of Eq. (70), renormalized and rapidity-renormalized. This makes explicit the division of labor:

SCET fixes the operator and evolution, the deuteron model supplies its matrix element. (71)

9 Leading-twist spin-1 TMD inventory

The TMD inventory is the forward limit of the GTMD helicity matrices in Sec. 5. The correct bookkeeping is species-first at the operator level and channel-first at the observable level. For each named TMD the registry must know which operator projection generated it:

Species	Leading operator projection	Parton polarization information
$q, \bar{q}$	$\Phi^{[\gamma^+]}$	quark unpolarized density
$q, \bar{q}$	$\Phi^{[\gamma^+ \gamma_5]}$	quark helicity density
$q, \bar{q}$	$\Phi^{[i\sigma^{i+} \gamma_5]}$	quark transverse-spin/chiral-odd density
g	$\Gamma^{ij} \delta_{Tij}$	unpolarized gluon density
g	$i\epsilon_T^{ij} \Gamma_{ij}$	circularly polarized gluon density
g	$\Gamma^{(ij)} - \frac{1}{2} \delta_T^{ij} \Gamma_k^k$	linearly polarized/double-helicity-flip gluon density

The spin-1 target projectors U, L, T, LL, LT, TT are then applied to the deuteron helicity indices. This construction lets the full set of polarized quark, antiquark, and gluon TMDs be generated systematically rather than by writing unrelated ansatz functions one by one.

A spin-1 target has the usual spin-1/2-like TMDs plus additional tensor-polarized TMDs. The quark sector has ten leading-twist tensor-polarized TMDs in addition to the spin-1/2-like leading-twist functions when time-reversal constraints are not imposed before specifying the gauge link [7, 8]. A practical inventory is shown in Table 2. The table is a model-organizing inventory, not a convention-independent decomposition; signs and prefactors must be fixed by the chosen correlator convention.

Table 2: Leading-twist quark TMD inventory for a spin-1 target. The first three rows are spin-1/2-like target polarizations; the last three rows are genuinely tensor-polarized target channels. Naive time-reversal labels are gauge-link dependent and should be assigned only after a process convention is chosen.

Target channel	Typical functions	Comments
U	$f_1, h_1^\perp$	Unpolarized and Boer–Mulders-type structures.
L	$g_{1L}, h_{1L}^\perp$	Vector-longitudinal target polarization.
T	$f_{1T}^\perp, g_{1T}, h_1, h_{1T}^\perp$	Vector-transverse target polarization.
LL	$f_{1LL}, h_{1LL}^\perp$	Longitudinal tensor polarization; f_{1LL} integrates to the b_1 tensor PDF up to convention factors.
LT	$f_{1LT}, g_{1LT}, h_{1LT}, h_{1LT}^\perp$	Mixed longitudinal-transverse tensor polarization.
TT	$f_{1TT}, g_{1TT}, h_{1TT}, h_{1TT}^\perp$	Transverse tensor polarization.

For the leading-twist, T-even gluon sector, we retain the notation used by Xie, Chen, and Lu for comparison [1]. Their T-even spin-1 gluon inventory is

$$\begin{aligned}
& f_1^g, \quad h_1^{\perp g}, \quad g_1^g, \quad g_{1T}^g, \\
& f_{1LL}^g, \quad f_{1LT}^g, \quad f_{1TT}^g, \quad h_{1LL}^{\perp g}, \quad h_{1LT}^{\perp g}, \quad h_{1TT}^{\perp g}, \\
& h_{1TT}^g, \quad h_{1TT}^{\perp g}, \quad h_{1TT}^{\perp\perp g}.
\end{aligned} \tag{72}$$

The function often described as gluon transversity or double-helicity-flip gluon structure is special because it exists for $\text{spin} \geq 1$ targets but has no spin-1/2 nucleon analogue [3, 8]. In this framework it should be treated as a null-test or exotic/coherent diagnostic, not as the sole organizing target of the model.

10 Tensor-basis construction and projectors

Rather than hard-code a decomposition from a single paper, we construct a transverse tensor basis and then map it to the notation of any chosen reference. For a correlator with two transverse gluon indices, write

$$\tilde{\Gamma}^{ij}(x, \mathbf{b}_T; S, T) = \sum_A \mathbb{B}_A^{ij}(\hat{\mathbf{b}}_T, S, T) \tilde{F}_A(x, b_T; \mu, \zeta), \tag{73}$$

where A runs over all allowed leading-power structures. The basis tensors $\mathbb{B}_A^{ij}$ are built from

$$\delta_T^{ij}, \quad \epsilon_T^{ij}, \quad \hat{b}^i, \quad S_T^i, \quad S_{LT}^i, \quad S_{TT}^{ij}, \quad (74)$$

plus the scalar polarizations S_L and S_{LL} . The symmetric traceless rank- r tensors generated by $\hat{\mathbf{b}}_T$ are

$$\tau_b^{ij} = \hat{b}^i \hat{b}^j - \frac{1}{2} \delta_T^{ij}, \quad (75)$$

$$\tau_b^{ijk} = \hat{b}^i \hat{b}^j \hat{b}^k - \frac{1}{4} \left(\delta_T^{ij} \hat{b}^k + \delta_T^{ik} \hat{b}^j + \delta_T^{jk} \hat{b}^i \right), \quad (76)$$

with higher ranks defined by symmetric-traceless projection. For k_T -space replace $\hat{\mathbf{b}}_T$ by $\hat{\mathbf{k}}_T$ and include powers of k_T/M_D .

The basis must satisfy Hermiticity, parity, leading-power transverse-rank counting, and the chosen naive time-reversal sector. Once a basis is chosen, define the Gram matrix

$$G_{AB} = \mathbb{B}_A^{ij} \mathbb{B}_{B,ij}, \quad (77)$$

with sums over transverse indices and, if necessary, averages over azimuthal angles and polarization tensors. The projectors are

$$\mathbb{P}_{A,ij} = \sum_B (G^{-1})_{AB} \mathbb{B}_{B,ij}, \quad \tilde{F}_A = \mathbb{P}_{A,ij} \tilde{\Gamma}^{ij}. \quad (78)$$

This projector construction is the safest way to avoid sign and normalization errors. The implementation should include a symbolic or numerical test:

$$\mathbb{P}_{A,ij} \mathbb{B}_B^{ij} = \delta_{AB}. \quad (79)$$

For quark TMDs the same construction applies after replacing the free gluon-index pair (i, j) by the Dirac projections

$$\Phi^{[\gamma^+]}, \quad \Phi^{[\gamma^+ \gamma_5]}, \quad \Phi^{[i\sigma^{i+} \gamma_5]}. \quad (80)$$

11 TMD convolution from the Wigner density

The TMD convolution is not an independent postulate. It is the $\Delta_T = 0$ reduction of the GTMD convolution in Eq. (45). This matters because the same nuclear kernel must also reproduce the GPD/form-factor marginal. In practice, the model should implement the GTMD-level convolution first and expose the TMD, GPD, and PDF reductions as views of the same parent object. The formulas below are the forward TMD specialization, written explicitly because they are the ones used in SCET/TMD phenomenology.

11.1 Momentum-space convolution

The leading nucleonic TMD convolution is

$$F_{a/D,I}^{\text{IA}}(x, \mathbf{k}_T; \mu, \zeta) = \sum_{N=p,n} \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} W_I^N(y, \mathbf{p}_T)$$

$$\times F_{a/N} \left(\frac{x}{y}, \mathbf{k}_T - \frac{x}{y} \mathbf{p}_T; \mu, \zeta_N \right) \mathcal{J}_T(y, \mathbf{p}_T) + \delta F_{a/D,I}^{\text{spin transfer}}. \quad (81)$$

Here $I = U, L, T, LL, LT, TT$ labels the deuteron polarization channel. The transverse shift is the leading kinematic boost from nucleon transverse motion to deuteron transverse momentum. The term $\delta F_{a/D,I}^{\text{spin transfer}}$ denotes effects in which the struck nucleon's polarization density matrix induces nucleon polarized TMDs. For a complete model, W_I^N must be generalized from a scalar density to a nucleon spin-density matrix,

$$W_{I;\lambda'_N \lambda_N}^N(y, \mathbf{p}_T), \quad (82)$$

so that vector-polarized nucleon TMDs can contribute to tensor-polarized deuteron observables through spin recoupling.

Equation (81) is the model's basic bridge between nuclear dynamics and partonic transverse momentum. It implies the collinear convolution after integrating over $\mathbf{k}_T$:

$$\int d^2 \mathbf{k}_T F_{a/D,I}^{\text{IA}}(x, \mathbf{k}_T) = \sum_N \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} W_I^N(y, \mathbf{p}_T) f_{a/N} \left(\frac{x}{y} \right) \mathcal{J}_T(y, \mathbf{p}_T), \quad (83)$$

provided the nucleon TMD integrates to the nucleon PDF.

11.2 b_T -space convolution

Fourier-transforming Eq. (81) gives a cleaner product form:

$$\begin{aligned} \tilde{F}_{a/D,I}^{\text{IA}}(x, \mathbf{b}_T; \mu, \zeta) &= \sum_N \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} W_I^N(y, \mathbf{p}_T) e^{i(x/y) \mathbf{b}_T \cdot \mathbf{p}_T} \\ &\times \tilde{F}_{a/N} \left(\frac{x}{y}, \mathbf{b}_T; \mu, \zeta_N \right) \mathcal{J}_T(y, \mathbf{p}_T) + \tilde{F}_{a/D,I}^{\text{spin transfer}}. \end{aligned} \quad (84)$$

This form is preferred for SCET evolution because evolution is multiplicative in b_T space. The exponential phase shows explicitly that the nuclear transverse-momentum distribution broadens the partonic TMD.

The full model is

$$\tilde{F}_{a/D,I} = \tilde{F}_{a/D,I}^{\text{IA}} + \tilde{F}_{a/D,I}^{\text{coh}} + \tilde{F}_{a/D,I}^{\text{exch}} + \tilde{F}_{a/D,I}^{\text{nonN}}. \quad (85)$$

At moderate and large x , IA should dominate many quark channels. At small x , coherent and shadowing terms can become important. For gluon double-helicity-flip channels, the one-body nucleon baseline may vanish or be strongly suppressed, making coh and nonN central rather than corrective.

12 Small- b_T OPE and matching

For $b_T \ll 1/\Lambda_{\text{QCD}}$, a TMD can be matched onto collinear PDFs:

$$\tilde{F}_{a/D,I}(x, b_T; \mu, \zeta) = \sum_{b=q,\bar{q},g} \int_x^1 \frac{dz}{z} C_{a \leftarrow b}^{[J,I]} \left(\frac{x}{z}, b_T; \mu, \zeta \right) f_{b/D,I}^{[J]}(z; \mu) + \mathcal{O}(b_T \Lambda_{\text{QCD}}). \quad (86)$$

The label J denotes the partonic polarization/Dirac/Lorentz channel, and I denotes target polarization. The perturbative coefficients know about the partonic operator; they do not know about deuteron nuclear binding except through the collinear matrix element.

At tree level, for the unpolarized quark and gluon channels,

$$C_{q \leftarrow q}^{(0)}(z) = \delta(1 - z), \quad C_{g \leftarrow g}^{(0)}(z) = \delta(1 - z), \quad (87)$$

with other channels vanishing. For tensor-polarized $f_{1LL}^{q,g}$, the same tree-level coefficient multiplies the corresponding tensor-polarized collinear PDF, once conventions are fixed:

$$\tilde{F}_{f_{1LL}}^{a/D}(x, b_T; \mu, \zeta) = f_{1LL}^{a/D}(x; \mu) + \mathcal{O}(\alpha_s, b_T \Lambda_{\text{QCD}}) \quad (88)$$

if f_{1LL} is normalized as the collinear integral of the TMD. In the helicity-difference convention,

$$\delta_T \tilde{F}_{a/D}(x, b_T; \mu, \zeta) = \delta_T f_{a/D}(x; \mu) + \mathcal{O}(\alpha_s, b_T \Lambda_{\text{QCD}}). \quad (89)$$

Spin and tensor-rank determine whether a TMD has a nonzero collinear integral. A rank- $r > 0$ angular structure vanishes under unweighted $\mathbf{k}_T$ integration but can have nonzero transverse moments. Define

$$F^{(n)}(x) = \int d^2 \mathbf{k}_T \left(\frac{k_T^2}{2M_D^2} \right)^n F(x, k_T^2). \quad (90)$$

The model should classify each TMD as one of:

- (i) nonzero collinear limit, directly constrained by PDFs or b_1 ;
- (ii) zero unweighted integral but nonzero transverse moment;
- (iii) perturbatively generated small- b_T tail with no independent collinear PDF;
- (iv) genuinely non-nucleonic/coherent boundary function.

For singlet channels, collinear evolution mixes quark singlet and gluon tensor PDFs when allowed by quantum numbers. The unpolarized and LL tensor-polarized singlet channels should use DGLAP mixing matrices analogous to the unpolarized singlet case, with tensor-polarized boundary distributions replacing unpolarized PDFs. Special double-helicity-flip gluon distributions require separate treatment because no spin-1/2 nucleon quark analogue exists.

13 TMD evolution

The renormalized TMDs obey

$$\frac{d}{d \ln \mu} \ln \tilde{F}_{a/D,I}(x, b_T; \mu, \zeta) = \gamma_F^a(\alpha_s(\mu), \zeta/\mu^2), \quad (91)$$

$$\frac{d}{d \ln \sqrt{\zeta}} \ln \tilde{F}_{a/D,I}(x, b_T; \mu, \zeta) = -\mathcal{D}_a(b_T; \mu), \quad (92)$$

with

$$\frac{d}{d \ln \mu} \mathcal{D}_a(b_T; \mu) = \Gamma_{\text{cusp}}^a(\alpha_s). \quad (93)$$

Here $a = q$ or g denotes the partonic color representation. At leading power the Collins–Soper kernel is independent of target polarization; the polarization dependence resides in the boundary functions and collinear matching.

A practical evolution solution from (μ_0, ζ_0) to (μ, ζ) is

$$\tilde{F}_{a/D,I}(x, b_T; \mu, \zeta) = R_a(b_T; \mu, \zeta; \mu_0, \zeta_0) \tilde{F}_{a/D,I}(x, b_T; \mu_0, \zeta_0), \quad (94)$$

where R_a is universal for a given parton representation and TMD scheme. In a b_* implementation,

$$b_* = \frac{b_T}{\sqrt{1 + b_T^2/b_{\max}^2}}, \quad \mu_b = \frac{2e^{-\gamma_E}}{b_*}, \quad (95)$$

with

$$R_a = \exp[-S_{\text{pert}}^a(Q, \mu_b, b_*) - S_{\text{NP}}^a(x, b_T, Q)]. \quad (96)$$

The nonperturbative Collins–Soper term may initially be taken polarization independent for a minimal model,

$$S_{\text{NP}}^a(x, b_T, Q) = g_K^a(b_T) \ln \frac{Q}{Q_0} + g_{a/D,I}(x, b_T), \quad (97)$$

then generalized if data demand target-polarization dependence. Because g_K^a is universal within a representation, polarization dependence in the CS kernel should be introduced only with strong justification.

14 Model ansatz over phase space

The boundary model at Q_0 is constructed in b_T space:

$$\tilde{F}_{a/D,I}(x, b_T; Q_0) = \tilde{F}_{a/D,I}^{\text{IA}}(x, b_T; Q_0) + \tilde{F}_{a/D,I}^{\text{coh}}(x, b_T; Q_0) + \tilde{F}_{a/D,I}^{\text{exch}}(x, b_T; Q_0) + \tilde{F}_{a/D,I}^{\text{nonN}}(x, b_T; Q_0). \quad (98)$$

The impulse term is given by Eq. (84). The remaining terms are modeled with constrained functional forms and fitted to inclusive and semi-inclusive data as available.

A flexible ratio model is useful for tensor-polarized functions:

$$\tilde{F}_{a/D,I}(x, b_T; Q_0) = R_{a,I}(x, b_T; Q_0) \tilde{F}_{a/D,U}(x, b_T; Q_0), \quad (99)$$

where

$$R_{a,I}(x, b_T) = N_{a,I} x^{\alpha_{a,I}} (1-x)^{\beta_{a,I}} P_{a,I}(x) \left(\frac{M_D b_T}{2} \right)^{r_I} \exp[-\lambda_{a,I} b_T^2]. \quad (100)$$

Here r_I is the transverse rank of the relevant tensor structure and $P_{a,I}(x)$ can be a low-order Bernstein polynomial or spline. Equation (100) is not the final physics model; it is a controllable starting point that enforces regularity at $b_T = 0$ and $x \rightarrow 0, 1$.

For a global phase-space model, the additive components can be turned on by smooth profiles, for example

$$\tilde{F}_{a/D,I}^{\text{coh}}(x, b_T; Q_0) = w_{\text{coh}}(x) \mathcal{A}_{a,I}^{\text{coh}}(x, b_T), \quad w_{\text{coh}}(x) = \frac{1}{1 + (x/x_s)^\eta}. \quad (101)$$

This allows the same code to describe moderate- x nucleonic physics and low- x coherent physics without claiming that the impulse approximation works everywhere.

15 Fourier–Bessel transforms and transverse rank

Theory and evolution are most naturally expressed in b_T space, while many model plots and observables are expressed in k_T space. A rank- r azimuthal tensor structure transforms with a Bessel function J_r :

$$F^{(r)}(x, k_T) = \int_0^\infty \frac{b_T db_T}{2\pi} J_r(b_T k_T) \text{ilde} F^{(r)}(x, b_T) \times \mathcal{N}_r, \quad (102)$$

where $\mathcal{N}_r$ depends on the convention used to factor powers of M_D , b_T , or k_T from the tensor basis. A model implementation should not use a single scalar Fourier transform for all TMDs. It should store each function's transverse rank and transform convention.

The inverse relation is similarly rank dependent:

$$\tilde{F}^{(r)}(x, b_T) = 2\pi \int_0^\infty k_T dk_T J_r(b_T k_T) F^{(r)}(x, k_T) \times \mathcal{N}_r^{-1}. \quad (103)$$

The unweighted collinear integral probes only the rank-zero piece:

$$\int d^2 \mathbf{k}_T F^{(r)}(x, k_T) = 0 \quad \text{for } r > 0 \quad (104)$$

unless the tensor basis has been defined so that the rank factors are already contracted into a scalar before integration. This is another reason to keep a machine-readable TMD registry.

16 SCET factorization for observables

16.1 Drell–Yan

For Drell–Yan at $q_T \ll Q$, the leading-power factorized cross section has the schematic form

$$\begin{aligned} \frac{d\sigma^{hD \rightarrow \ell\ell X}}{dQ^2 dY d^2 \mathbf{q}_T} &= \sigma_0 H_{q\bar{q}}(Q, \mu) \sum_q e_q^2 \int \frac{d^2 \mathbf{b}_T}{(2\pi)^2} e^{i\mathbf{b}_T \cdot \mathbf{q}_T} \\ &\times \tilde{F}_{q/h}(x_h, b_T; \mu, \zeta_h) \tilde{F}_{\bar{q}/D, I}(x_D, b_T; \mu, \zeta_D) + (q \leftrightarrow \bar{q}) + Y + \mathcal{O}(q_T/Q). \end{aligned} \quad (105)$$

The deuteron polarization label I determines which angular modulation and tensor asymmetry is projected. Drell–Yan is especially valuable for antiquark tensor polarization. Gluon transversity-like effects may enter specialized double-helicity-flip observables, but such channels currently lack direct data constraints.

16.2 SIDIS

For SIDIS $\ell D \rightarrow \ell' h X$ in the TMD region $P_{hT} \ll Q$,

$$\begin{aligned} \frac{d\sigma}{dx_B dz_h dQ^2 d^2 \mathbf{P}_{hT}} &= \sigma_0 H_{\text{SIDIS}}(Q, \mu) \sum_q e_q^2 \int \frac{d^2 \mathbf{b}_T}{(2\pi)^2} e^{i\mathbf{b}_T \cdot \mathbf{P}_{hT}/z_h} \\ &\times \tilde{F}_{q/D, I}(x_B, b_T; \mu, \zeta_D) \tilde{D}_{h/q}(z_h, b_T; \mu, \zeta_h) + Y_{\text{SIDIS}} + \mathcal{O}(P_{hT}/Q). \end{aligned} \quad (106)$$

The spin-1 SIDIS cross section contains many tensor-dependent structure functions [6, 7, 11]. In practice, early constraints are likely to come from LL tensor observables such as f_{1LL} and related modulations.

16.3 Inclusive DIS limit

Integrating SIDIS over the observed hadron and transverse momentum, or applying inclusive DIS factorization directly, gives the collinear structure functions. The model must satisfy

$$\int d^2\mathbf{k}_T \delta_T f_{q/D}(x, \mathbf{k}_T; Q) = \delta_T q_D(x, Q), \quad (107)$$

which inserted into Eq. (23) reproduces b_1 . This is the first validation test for the TMD model.

16.4 Gluon-sensitive observables

Gluon TMDs can be accessed in gluon-sensitive channels such as heavy-quark-pair production, dijet production, quarkonium production with appropriate treatment of NRQCD assumptions, or low- x final states at an EIC. The generic tensor structure is

$$d\sigma_{g\text{-sens.}} \sim H_{ij,kl}(Q, \mu) \int \frac{d^2\mathbf{b}_T}{(2\pi)^2} e^{i\mathbf{b}_T \cdot \mathbf{q}_T} \tilde{\Gamma}_{g/D,I}^{ij}(x_D, b_T; \mu, \zeta_D) \mathcal{F}_{\text{other}}^{kl}(x, b_T; \mu, \zeta) + Y. \quad (108)$$

For colored final states, possible Glauber exchanges and factorization-breaking effects must be considered. The model should mark gluon-sensitive channels by the status of their factorization theorem: established, assumed, or exploratory.

17 Matching low, moderate, and high transverse momentum

A full phase-space description requires the standard $W + Y$ construction:

$$d\sigma = d\sigma_W + d\sigma_Y, \quad (109)$$

where $d\sigma_W$ is the resummed TMD expression valid at $q_T \ll Q$, and

$$d\sigma_Y = d\sigma_{\text{FO}} - [d\sigma_W]_{\text{expanded to same order}} \quad (110)$$

restores the fixed-order behavior at $q_T \sim Q$. The TMD model itself should not be forced to describe the entire high- k_T tail nonperturbatively. The tail is constrained by perturbative matching and by the Y term in observables.

At the distribution level, the model should enforce:

1. small- b_T consistency with the OPE in Eq. (86);
2. large- b_T nonperturbative behavior controlled by the Wigner/nuclear model and the CS kernel;
3. smooth matching between resummed and fixed-order regions at the observable level.

18 Positivity, sum rules, and consistency tests

The framework must enforce several classes of constraints.

18.1 Density-matrix positivity

The deuteron spin-density matrix must satisfy

$$\rho = \rho^\dagger, \quad \text{Tr} \rho = 1, \quad v^\dagger \rho v \geq 0 \quad \text{for all } v \in \mathbb{C}^3. \quad (111)$$

The LF nucleon density matrix obtained from wave-function overlaps automatically satisfies positivity if no components are dropped. If phenomenological tensor densities are introduced independently, they must be checked by reconstructing the full 3×3 matrix.

18.2 Partonic helicity-density positivity

For each (x, k_T) and process gauge-link structure, the quark or gluon TMD correlator defines a parton-target helicity matrix. Positivity of this matrix gives inequalities among TMDs. For gluon TMDs of spin ≤ 1 , detailed positivity bounds are known and should be used as constraints or validation tests [10]. A robust implementation should impose semidefinite positivity numerically rather than relying only on a subset of analytic inequalities.

18.3 Collinear and tensor sum rules

The unpolarized deuteron distributions should satisfy baryon-number and momentum sum rules within the chosen nuclear convention. The tensor-polarized quark distributions must reproduce b_1 through Eq. (23). Additional tensor-polarized sum rules depend on assumptions about sea-quark tensor polarization and the small- x behavior; they should be implemented as optional constraints rather than hard-coded identities unless their hypotheses are satisfied.

18.4 Evolution consistency

The following checks are mandatory:

$$\frac{d}{d \ln \mu} \frac{d}{d \ln \sqrt{\zeta}} \ln \tilde{F} = \frac{d}{d \ln \sqrt{\zeta}} \frac{d}{d \ln \mu} \ln \tilde{F}, \quad (112)$$

$$\frac{d}{d \ln \mu} \mathcal{D}_a(b_T; \mu) = \Gamma_{\text{cusp}}^a(\alpha_s), \quad (113)$$

$$\int d^2 \mathbf{k}_T F_{a/D,I}(x, k_T; Q) = f_{a/D,I}(x; Q) \quad \text{for rank-zero collinear channels.} \quad (114)$$

19 Numerical infrastructure

The framework should be implemented as a modular code base. The minimal internal registry for a TMD should contain

$$\begin{aligned}
\text{name :} & f_1, f_{1LL}, h_{1TT}^g, \dots \\
\text{species :} & q, \bar{q}, g \\
\text{parent_GTMD :} & \mathbb{W}_{\Lambda'\Lambda}^{a[J]}(x, \mathbf{k}_T, \mathbf{\Delta}_T; \eta) \\
\text{operator_projection :} & \gamma^+, \gamma^+ \gamma_5, i\sigma^{i+} \gamma_5, \Gamma_g^{ij} \\
\text{target_channel :} & U, L, T, LL, LT, TT \\
\text{parton_polarization :} & \text{unpolarized, helicity, transversity/linear, etc.} \\
\text{rank :} & r = 0, 1, 2, \dots \\
\text{gauge_link :} & [+ , +], [- , -], [+ , -], [- , +], \dots \\
\text{marginals :} & \text{TMD/GPD/PDF/form-factor reductions available} \\
\text{collinear_limit :} & \text{yes/no/moment only} \\
\text{matching_status :} & \text{known/tree/derived/modeled} \\
\text{positivity_block :} & \text{matrix block used for bounds.}
\end{aligned} \tag{115}$$

A natural module structure is:

- M1. Kinematics and conventions:** light-front components, Fourier transforms, tensor bases.
- M2. Deuteron wave functions:** AV18, CD-Bonn, chiral EFT; LF mapping and spin rotations.
- M3. Wigner/spectral densities:** helicity density matrices, U, L, T, LL, LT, TT projections.
- M4. GTMD parent layer:** zero-skewness quark and gluon GTMD objects, Wigner transforms, TMD/GPD/PDF marginal methods.
- M5. Nucleon inputs:** proton/neutron PDFs, TMDs, and where available GTMD/GPD models, plus fragmentation functions for SIDIS.
- M6. Nuclear mechanisms:** impulse, coherent/shadowing, exchange currents, non-nucleonic terms.
- M7. TMD evolution:** perturbative Sudakov factors, nonperturbative CS kernel, b_* prescription.
- M8. Observable kernels:** DIS, SIDIS, DY, and gluon-sensitive processes.
- M9. Constraints:** b_1 , positivity, sum rules, small- b_T matching, uncertainty priors.

20 Comparison with the gluon spectator model

The spectator calculation of Xie, Chen, and Lu provides a useful comparison target for the leading-twist, T-even gluon subset [1]. The comparison should be performed only after a convention dictionary is built. For each gluon TMD in Eq. (72), record:

1. tensor-polarization convention and sign;
2. mass normalization factors M_D , k_T/M_D , and transverse moments;
3. gauge-link convention or absence thereof in the model calculation;

4. input scale, PDF set, and running/evolution treatment;
5. Fourier-transform convention between b_T and k_T space.

The point of comparison is not to force agreement. Differences should be classified as:

$$\Delta F = \Delta F_{\text{convention}} + \Delta F_{\text{nuclear model}} + \Delta F_{\text{perturbative tail}} + \Delta F_{\text{evolution}} + \Delta F_{\text{coherent/nonN}}. \quad (116)$$

The spectator model encodes low-scale dynamics through an effective deuteron–gluon–spectator vertex and spectral mass distribution. The Wigner+SCET framework encodes low-scale dynamics through nuclear wave functions and constrained nonperturbative matrix elements, while imposing operator definitions and evolution. Agreement in qualitative shape would be encouraging; disagreement in gluon-transversity-like channels may be physically informative.

21 Development roadmap

The framework should be developed in stages:

Stage 0: GTMD parent layer and consistency algebra

1. Choose the zero-skewness GTMD convention, Wilson-line labels, Fourier-transform signs, and Δ_T recoil convention.
2. Implement quark and gluon parent correlator containers with target-helicity matrices.
3. Implement the reductions $\mathbb{W} \rightarrow \text{TMD}$, $\mathbb{W} \rightarrow \text{GPD}$, $\mathbb{W} \rightarrow \text{PDF}$, and local-current/form-factor moments.
4. Build symbolic/numerical projectors from GTMD helicity matrices to named spin-1 TMDs.

Stage 1: Inclusive tensor PDFs

1. Implement LF deuteron wave functions and helicity density matrices.
2. Compute $\delta_T \rho^N(y, p_T)$ and $\delta_T q_D(x)$ in impulse approximation.
3. Reproduce Eq. (23) and compare with existing b_1 data.
4. Add flexible coherent/exchange/non-nucleonic components if impulse approximation is insufficient.

Stage 2: Rank-zero and low-rank TMDs

1. Build $f_1^{a/D}$ and $f_{1LL}^{a/D}$ in b_T and k_T space.
2. Check collinear integrals and tensor normalization.
3. Add SIDIS tensor observables most directly related to f_{1LL} and $h_{1LL}^\perp$.

Stage 3: Full spin-1 quark, antiquark, and gluon registry

1. Add all leading-twist quark TMD channels in Table 2.
2. Add the T-even gluon channels in Eq. (72); later extend to T-odd/gauge-link-dependent channels.
3. Implement positivity matrices for quark and gluon sectors.

Stage 4: SCET evolution and process observables

1. Implement μ, ζ evolution in a fixed scheme.
2. Add the $W + Y$ matching for SIDIS and DY.
3. Add candidate gluon-sensitive observables with explicit factorization-status labels.

Stage 5: Global inference

1. Fit or constrain the model using b_1 , SIDIS tensor observables, unpolarized deuteron/nuclear PDFs, and nucleon TMD information.
2. Propagate wave-function, nuclear-model, TMD-evolution, and small- x uncertainties.
3. Produce predictions for tensor-polarized SIDIS, DY, and gluon-sensitive EIC observables.

22 Open derivations and near-term tasks

This first draft intentionally leaves several calculations as explicit tasks rather than hiding them in assumptions:

- T1.** Fix the zero-skewness GTMD convention, including Fourier signs, gauge-link labels, Δ_T recoil factors, and the mapping between partonic Wigner impact parameter and TMD-evolution impact parameter.
- T2.** Prove explicitly that the chosen GTMD convolution reduces to the TMD convolution, the collinear PDF convolution, and the zero-skewness GPD/form-factor convolution with one common nuclear kernel.
- T3.** Derive the exact LF Jacobian and spin-rotation convention used in Eq. (53) for the selected wave-function implementation.
- T4.** Work out the full nucleon spin-density convolution, including how vector-polarized nucleon TMDs feed tensor-polarized deuteron observables.
- T5.** Build the complete quark and gluon tensor-basis projector library and validate against Bacchetta–Mulders, Kumano–Song, Boer et al., and Xie–Chen–Lu conventions.
- T6.** Determine which tensor-polarized small- b_T matching coefficients are known, which are immediate by symmetry, and which require new one-loop calculations.
- T7.** Implement semidefinite positivity constraints for the full spin-1 quark and gluon helicity matrices.

- T8.** Decide the minimal coherent/shadowing model needed to describe the low- x behavior of b_1 and tensor-polarized sea distributions.
- T9.** Define process-specific Wilson-line labels for each planned observable and identify where Glauber/factorization-breaking issues may enter.

23 Conclusion

The framework proposed here is a hybrid of nuclear structure and QCD factorization. The deuteron is modeled through a light-front spin-density/Wigner distribution built from realistic wave functions and supplemented by coherent and non-nucleonic mechanisms. The partonic TMDs are defined through gauge-invariant, soft-subtracted QCD/SCET operators and evolved with Collins–Soper/TMD evolution. Inclusive DIS, especially b_1 , fixes the tensor-polarized quark normalization; SIDIS and DY provide the practical next constraints; gluon-sensitive observables and gluon transversity are included as important but less directly constrained sectors. This is the appropriate foundation for a serious spin-1 deuteron TMD model over phase space.

A Useful helicity combinations

For any helicity-dependent distribution F^Λ define

$$F_U = \frac{1}{3}(F^+ + F^0 + F^-), \quad (117)$$

$$F_L = \frac{1}{2}(F^+ - F^-), \quad (118)$$

$$\delta_T F = F^0 - \frac{1}{2}(F^+ + F^-). \quad (119)$$

These can be inverted as

$$F^0 = F_U + \frac{2}{3}\delta_T F, \quad (120)$$

$$F^+ = F_U + F_L - \frac{1}{3}\delta_T F, \quad (121)$$

$$F^- = F_U - F_L - \frac{1}{3}\delta_T F. \quad (122)$$

This normalization is often more transparent for comparing to b_1 than the S_{LL} normalization.

B Rank-dependent Bessel kernels

With the convention

$$\tilde{F}(\mathbf{b}_T) = \int d^2\mathbf{k}_T e^{i\mathbf{b}_T \cdot \mathbf{k}_T} F(\mathbf{k}_T), \quad (123)$$

rank-zero functions obey

$$\tilde{F}^{(0)}(b_T) = 2\pi \int_0^\infty k_T dk_T J_0(b_T k_T) F^{(0)}(k_T), \quad (124)$$

$$F^{(0)}(k_T) = \int_0^\infty \frac{b_T db_T}{2\pi} J_0(b_T k_T) \tilde{F}^{(0)}(b_T). \quad (125)$$

For rank-one structures proportional to $\hat{k}^i$ or k_T^i/M , the angular integral produces J_1 . For rank-two symmetric-traceless structures it produces J_2 . Higher-rank tensor structures follow analogously. Every TMD must store both its tensor rank and the mass/power convention used to make the scalar amplitude dimensionless.

C Minimal data and input list

The first implementation should include:

1. Deuteron wave functions: AV18 and CD-Bonn as baseline alternatives.
2. Proton and neutron PDFs: a modern PDF set, plus nuclear/deuteron corrections where needed.
3. Tensor DIS: HERMES b_1 and Jefferson Lab projected/available b_1 information.
4. Nucleon TMD inputs: unpolarized and polarized TMD parameterizations for proton/neutron where available.
5. Fragmentation functions for SIDIS.
6. TMD evolution kernel: one default nonperturbative Collins–Soper kernel, with uncertainty variations.

References

- [1] X. Xie, D.-Y. Chen, and Z. Lu, “Gluon TMDs for tensor polarized deuteron in a spectator model,” arXiv:2603.15224 [hep-ph] (2026).
- [2] P. Hoodbhoy, R. L. Jaffe, and A. Manohar, “Novel effects in deep inelastic scattering from spin-one hadrons,” Nucl. Phys. B **312**, 571–588 (1989).
- [3] R. L. Jaffe and A. Manohar, “Nuclear gluonometry,” Phys. Lett. B **223**, 218–224 (1989).
- [4] A. Airapetian et al. [HERMES Collaboration], “First measurement of the tensor structure function b_1 of the deuteron,” Phys. Rev. Lett. **95**, 242001 (2005), arXiv:hep-ex/0506018.
- [5] W. Cosyn, Y.-B. Dong, S. Kumano, and M. Sargsian, “Tensor-polarized structure function b_1 in the standard convolution description of the deuteron,” Phys. Rev. D **95**, 074036 (2017), arXiv:1702.05337 [hep-ph].
- [6] A. Bacchetta and P. J. Mulders, “Deep inelastic leptonproduction of spin-one hadrons,” Phys. Rev. D **62**, 114004 (2000), arXiv:hep-ph/0007120.
- [7] S. Kumano and Q.-T. Song, “Transverse-momentum-dependent parton distribution functions for spin-1 hadrons,” Phys. Rev. D **103**, 014025 (2021), arXiv:2011.08583 [hep-ph].
- [8] S. Kumano and Q.-T. Song, “Gluon transversity and TMDs for spin-1 hadrons,” arXiv:2201.04875 [hep-ph] (2022).

- [9] D. Boer, S. Cotogno, T. van Daal, P. J. Mulders, A. Signori, and Y.-J. Zhou, “Gluon and Wilson loop TMDs for hadrons of spin ≤ 1 ,” JHEP **10**, 013 (2016), arXiv:1607.01654 [hep-ph].
- [10] S. Cotogno, T. van Daal, and P. J. Mulders, “Positivity bounds on gluon TMDs for hadrons of spin ≤ 1 ,” JHEP **11**, 185 (2017), arXiv:1709.07827 [hep-ph].
- [11] J. Zhao, “Semi-inclusive deep inelastic scattering off a tensor-polarized deuteron,” JHEP **12**, 067 (2025), arXiv:2508.06134 [hep-ph].
- [12] C. W. Bauer, S. Fleming, D. Pirjol, and I. W. Stewart, “An effective field theory for collinear and soft gluons: Heavy to light decays,” Phys. Rev. D **63**, 114020 (2001), arXiv:hep-ph/0011336.
- [13] C. W. Bauer, D. Pirjol, and I. W. Stewart, “Soft-collinear factorization in effective field theory,” Phys. Rev. D **65**, 054022 (2002), arXiv:hep-ph/0109045.
- [14] T. Becher, A. Broggio, and A. Ferroglia, “Introduction to Soft-Collinear Effective Theory,” Lect. Notes Phys. **896**, 1–206 (2015), arXiv:1410.1892 [hep-ph].
- [15] J.-Y. Chiu, A. Jain, D. Neill, and I. Z. Rothstein, “The rapidity renormalization group,” Phys. Rev. Lett. **108**, 151601 (2012), arXiv:1104.0881 [hep-ph].
- [16] J.-Y. Chiu, A. Jain, D. Neill, and I. Z. Rothstein, “A formalism for the systematic treatment of rapidity logarithms in quantum field theory,” JHEP **05**, 084 (2012), arXiv:1202.0814 [hep-ph].
- [17] M. G. Echevarria, A. Idilbi, and I. Scimemi, “Factorization theorem for Drell-Yan at low q_T and transverse-momentum distributions on-the-light-cone,” JHEP **07**, 002 (2012), arXiv:1111.4996 [hep-ph].
- [18] M. G. Echevarria, A. Idilbi, and I. Scimemi, “Soft and collinear factorization and transverse momentum dependent parton distribution functions,” Phys. Lett. B **726**, 795–801 (2013), arXiv:1211.1947 [hep-ph].
- [19] J. Collins, *Foundations of Perturbative QCD*, Cambridge University Press (2011).
- [20] R. B. Wiringa, V. G. J. Stoks, and R. Schiavilla, “Accurate nucleon-nucleon potential with charge-independence breaking,” Phys. Rev. C **51**, 38–51 (1995), arXiv:nucl-th/9408016.
- [21] R. Machleidt, “The high-precision, charge-dependent Bonn nucleon-nucleon potential,” Phys. Rev. C **63**, 024001 (2001), arXiv:nucl-th/0006014.
- [22] A. V. Belitsky, X. Ji, and F. Yuan, “Quark imaging in the proton via quantum phase-space distributions,” Phys. Rev. D **69**, 074014 (2004), arXiv:hep-ph/0307383.
- [23] S. Meissner, A. Metz, and M. Schlegel, “Generalized parton correlation functions for a spin-1/2 hadron,” JHEP **08**, 056 (2009), arXiv:0906.5323 [hep-ph].
- [24] C. Lorcé and B. Pasquini, “Quark Wigner distributions and orbital angular momentum,” Phys. Rev. D **84**, 014015 (2011), arXiv:1106.0139 [hep-ph].
- [25] T. Neff and H. Feldmeier, “The Wigner function and short-range correlations in the deuteron,” arXiv:1610.04066 [nucl-th].